

SECTION 030100

CONCRETE REHABILITATION

PART 1 GENERAL

1.1 SUMMARY

- A. This section describes the patching of interior and exterior vertical and horizontal surfaces with a polymer modified, Portland cement repair mortar.

1.2 RELATED SECTIONS

- A. Section 033000 – Cast in Place Concrete

1.3 REFERENCES

- A. American Society for Testing and Materials
 - 1. ASTM C-78 – Test Method for Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading).
 - 2. ASTM C-109 – Test Method for Compressive Strength of Hydraulic Cement Mortars Using 2 in. or 50 mm. Cube Specimens.
 - 3. ASTM C-157 – Test Method for Length Change of Hardened Hydraulic Cement Mortar and Concrete.
 - 4. ASTM C-266 – Rate of Set with Gilmore Needle.

1.4 SUBMITTALS

- A. Submit under provisions of Section 013300 – Submittal Procedures
 - 1. Product Data: Manufacturer's literature to include surface preparation, application instructions, recommendations and storage and handling requirements.
 - 2. Test Data: Confirm compliance and performance with specified requirements.

1.5 QUALITY ASSURANCE

- A. Applicator Qualifications: Applicator must have prior experience applying specified product or similar products, or have manufacturer's representative on site ensuring that preparation and application are performed correctly.

B. Mockup:

- 1. Architect will select area for mockup.
- 2. Prior notice will be given to architect four days before mockups will be applied.
- 3. Architect must approve mockup before final product is applied.
- 4. At the architect's discretion, approved mockups may become incorporated into the final work.

1.6 DELIVERY, STORAGE & HANDLING

- A. Materials must be delivered in original, unopened containers with the manufacturer's labels including product name and batch numbers.

- B. Store material in a dry area, above ground. Protect cement from moisture and humidity.

1.7 ENVIRONMENTAL REQUIREMENTS

- A. Environmental Conditions: Do not apply material when temperature is below 7°C or when temperature is expected to fall below 7°C within 48 hours.
- B. Protection: Precautions should be taken to avoid damage to any surface near the work zone.

PART 2 PRODUCTS

2.1 MANUFACTURER

- A. Acceptable Manufacturer:
Saudi Vetonit Company Ltd.-Saveto
71 St. Second Industrial Estate Riyadh P.O.Box 52235 Riyadh 11563 Saudi Arabia Tel
+966 11 2653334, Fax: +966 11 265 0936, E-mail: info.Riyadh@saveto.com; Web:
www.saveto.com.

2.2 MATERIALS

- A. Cement based polymer modified fairing coat
 - 1. Vetorep CR523
 - 2. Description: single component polymer modified fibered cementitious fairing coat

2.3 MATERIAL PROPERTIES

- | | |
|---|-----------------------------------|
| A. Compressive Strength ASTM C109/109M: | > 20 N/mm ² @ 28 days. |
| B. Tensile Strength ASTM C190: | > 1.5 MPa @ 28 days. |
| C. Fresh Wet Density: | 1.8 kg/Ltr |
| D. Working life: | 40 minutes @ 35°C |
| E. Setting Time: | 2-3 hours |
| F. Coefficient of thermal expansion: | 10 to 12 x 10 ⁻⁶ / °C |
| G. VOC Content: | <10 gm < Liter |

PART 3 EXECUTION

3.1 EXAMINATION

- A. Examine areas to be repaired. Notify Engineer if surfaces are unacceptable. Do not begin surface preparation or application until unacceptable conditions are corrected.

3.2 SURFACE PREPARATION

- A. All surfaces should be clean, sound and uncontaminated prior to the application.
- B. The prepared area should be cleaned thoroughly by brush and/or compressed air.
- C. The prepared area should be soaked with water immediately prior to the application of Vetorep CR523.
- D. Areas to be repaired must be clean, sound, and free of dirt, oil, grease, laitance and other contaminants. Clean concrete with citrus-based concrete cleaner and stripper to remove grease.

3.3 MIXING

- A. Mix in an appropriate sized mechanical mixer or mix in a bucket using a drill and paddle mixer at low speed (400-600 RPM). First, pour water into the mixing container. Then, add powder material while continuing to mix. Mix to a uniform consistency for a minimum of three minutes and a maximum of five minutes.
- B. Mix only the amount of material that can be placed within working time. Do not re-temper material.
- C. Do not blend in excess liquid. Do not use any other admixtures or additives.

3.4 APPLICATION

- A. Mortar must be placed immediately upon completion of mixing.
- B. By using a steel trowel, the mixed mortar can be applied to the prepared substrates at up to 3 mm thicknesses. The applied mortar should be allowed to stiffen before troweling to a smooth finish.
- C. When material has reached proper set, shave, shape or mold with a steel trowel.

3.5 PROTECTION

- A. Protect horizontal surfaces from traffic until repair mortar compound has cured.

END OF SECTION